



DATA SCIENCE FIELD NOTES / 03A

BUSINESS INSIGHTS REPORT

Evidence-led findings, responsible interpretation and decisions for the next modelling phase

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PROGRAMME	AnalystLab Africa - Data Science Internship
PROJECT	Loan Approval Analysis and Feature Engineering
PHASE	Week 3 Completed analytical preparation

Educational portfolio report. Historical approval patterns do not constitute a production lending model or lending policy.

Executive summary

This report evaluates 614 historical loan applications to identify the factors most closely associated with approval and to prepare a responsible foundation for predictive modelling. Of the applications, 422 were approved and 192 were rejected, producing a historical approval rate of 68.73%.

Principal finding: applicants with positive credit history recorded a 79.05% approval rate, compared with 7.87% for applicants without positive credit history.

Portfolio measure	Verified result
Applications analysed	614
Approved / rejected	422 / 192
Historical approval rate	68.73%
Training / test observations	491 / 123
Engineered features	12

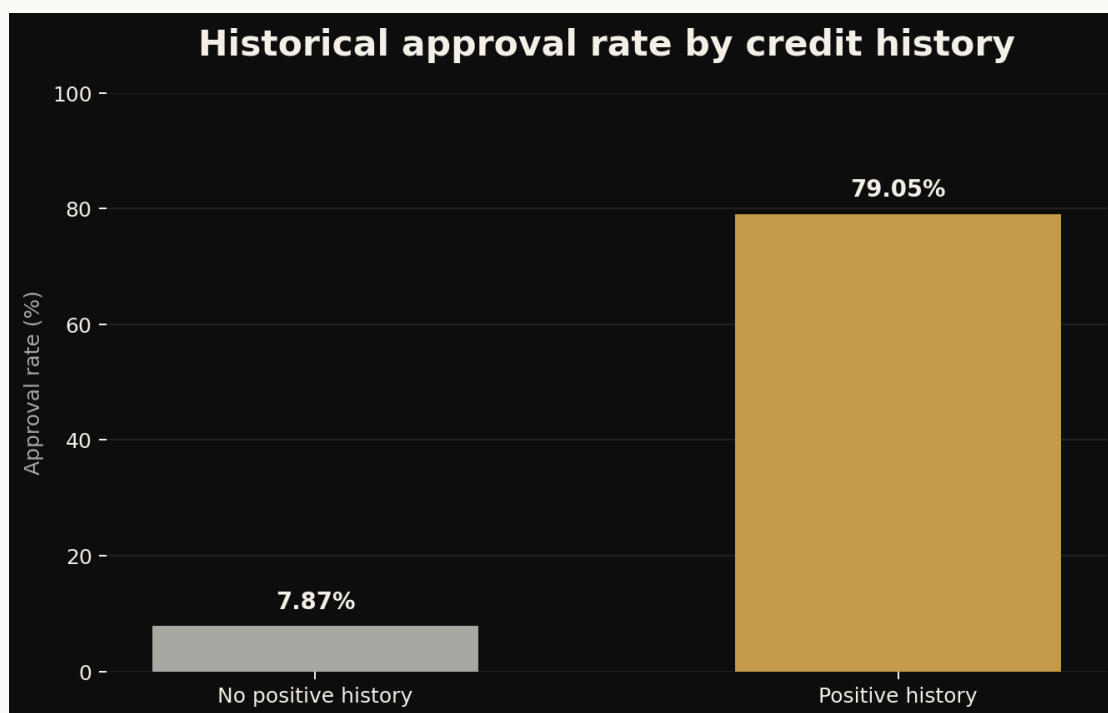


Figure 1. Historical approval rate by recorded credit-history status.

Business context and analytical approach

Lending institutions must balance access to credit with consistent risk assessment. This project examines whether applicant characteristics, household income, requested loan size, credit information and property area show meaningful relationships with historical approval decisions.

The workflow combines quality validation, exploratory visualisation, six hypothesis tests, feature engineering, responsible feature selection and leakage-safe preparation. It does not estimate repayment performance or actual default risk.

Credit history is the dominant observed factor

The relationship between credit history and approval is statistically significant (chi-square = 176.115; $p < 0.001$). Cramer's V of 0.536 is materially stronger than the other tested categorical association. Credit-history completeness and verification should therefore remain central to later modelling and data-quality controls.

Property area is a weaker secondary signal

Approval rates are 76.82% in semiurban areas, 65.84% in urban areas and 61.45% in rural areas. The association is statistically significant ($p = 0.002$), but Cramer's V is only 0.142. Geographic differences require investigation before they influence policy or operations.

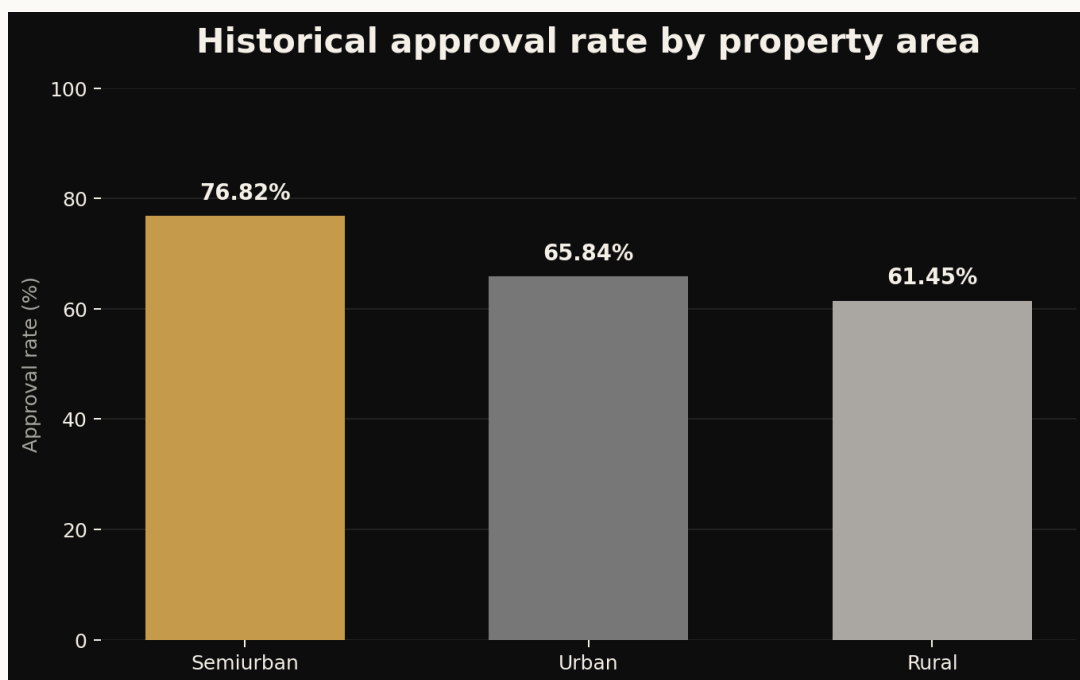


Figure 2. Property-area rates differ, but the association is much weaker than credit history.

What the financial variables reveal

Total household income and requested loan amount do not differ significantly between approved and rejected applications when assessed separately. Total income yields $p = 0.713$ and loan amount yields $p = 0.398$. These results do not make financial information irrelevant; they show that isolated raw values do not explain historical outcomes in this sample.

Total income and requested loan amount are strongly related (Spearman rho = 0.688; $p < 0.001$). Higher-income households generally request larger loans. Affordability should therefore be examined as a relationship among income, amount and repayment term rather than as an isolated threshold.

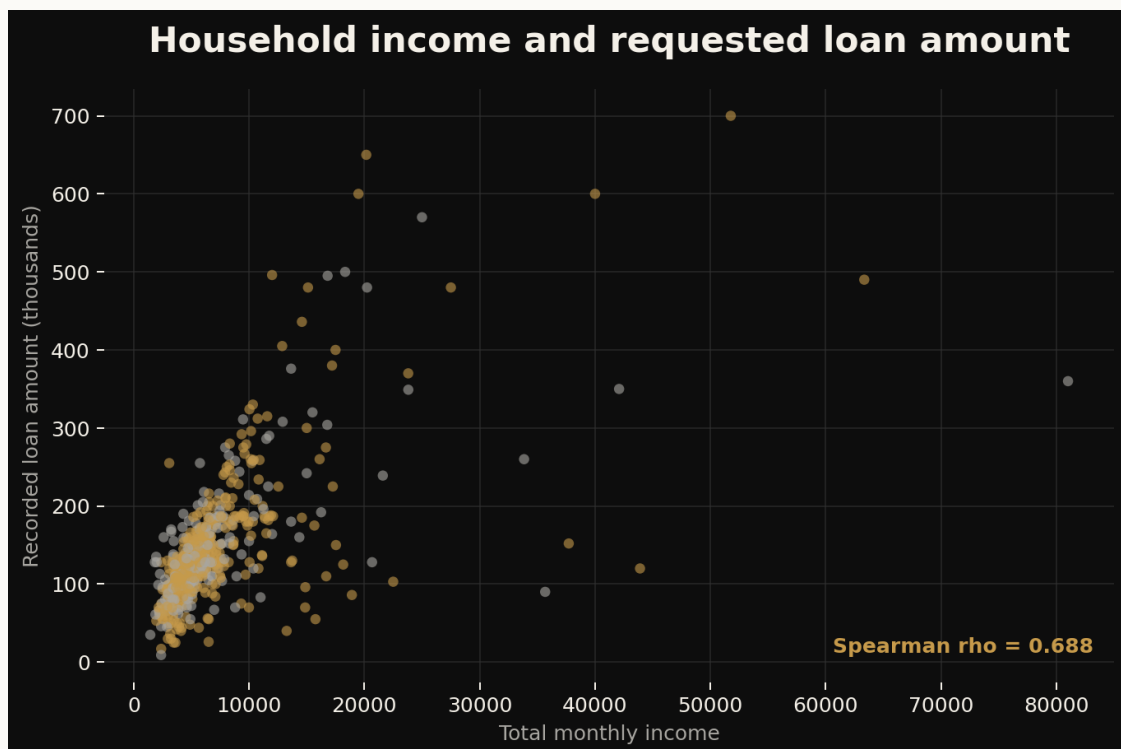


Figure 3. Household income and requested loan amount show a strong positive monotonic relationship.

Feature-engineering decisions

- Payment-income ratio combines approximate monthly principal, income and term into an affordability proxy.
- Coapplicant-income share describes dependence on a second income source.
- Log transformations reduce the influence of highly skewed financial values.
- Readable household, income and credit categories support communication but are controlled for redundancy in modelling.

Recommended actions

- Prioritise completeness and verification of credit-history information.
- Assess affordability through combined repayment and income indicators.
- Investigate geographic differences for product, branch-policy or data-collection effects.
- Monitor demographic fairness without using gender as a predictive input.
- Preserve the predetermined test partition and fit every transformation on training data only.

Limitations and responsible interpretation

- The dataset contains only 614 applications, so estimates for smaller groups may be unstable.
- Historical approval is the target, not repayment performance or default risk.
- Interest rates, existing debt, expenses and verified monthly instalments are unavailable.
- Previously imputed values may reduce natural variation.
- Application dates and applicant identifiers are absent, limiting temporal and repeat-applicant analysis.
- Observed associations may reflect historical institutional bias and do not establish causation.

Next steps

- Train a transparent baseline before comparing more complex classifiers.
- Use stratified cross-validation and evaluate precision, recall, F1, ROC-AUC and calibration.
- Audit subgroup errors and document threshold trade-offs.
- Keep human review, lending policy and compliance controls outside the model.

Conclusion

The Week 3 investigation transforms a cleaned application dataset into a statistically examined, feature-engineered and modelling-ready resource. Credit history is the clearest observed approval factor, while property area contributes a smaller signal and financial variables become more informative when combined into affordability-oriented representations.